4. Reference and assessment models

"Software Development Methodologies"
Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

CMMI (Capability Maturity Model Integration)

- Framework providing guidance for improving the processes of an organization and its ability to develop, manage and support the software products and services.
- Level of maturity is determined by the practice of diferente sets of sw development process activities.

https://www.youtube.com/watch?v=kF8sxDDoRns cmmi1

CMMI (Capability Maturity Model Integration)

- Levels are sequential and accumalative (e.g. being at Level X is expected to have elevated from Level X-1).
- Appraisal and training available by CMMI assessors.

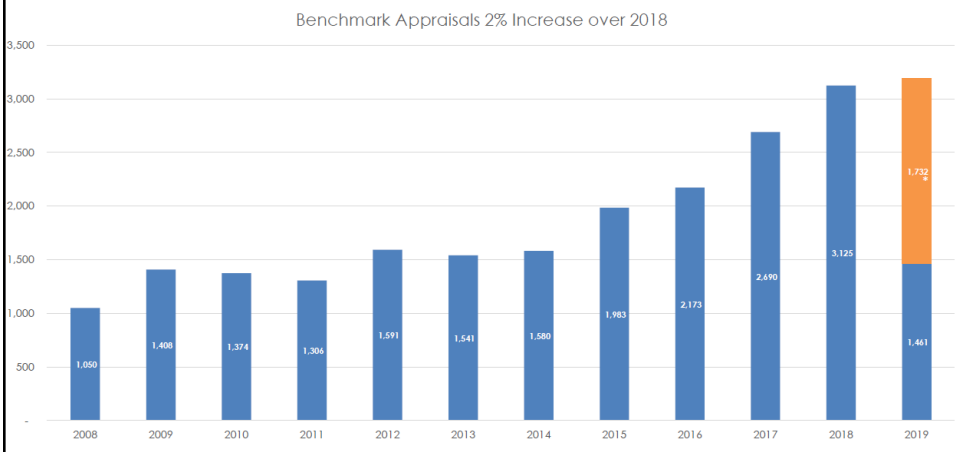
Who Uses CMMI?

Thousands of high-performing organizations have achieved sustainable business success through CMMI adoption and demonstrated their ability as capable business partners and suppliers.



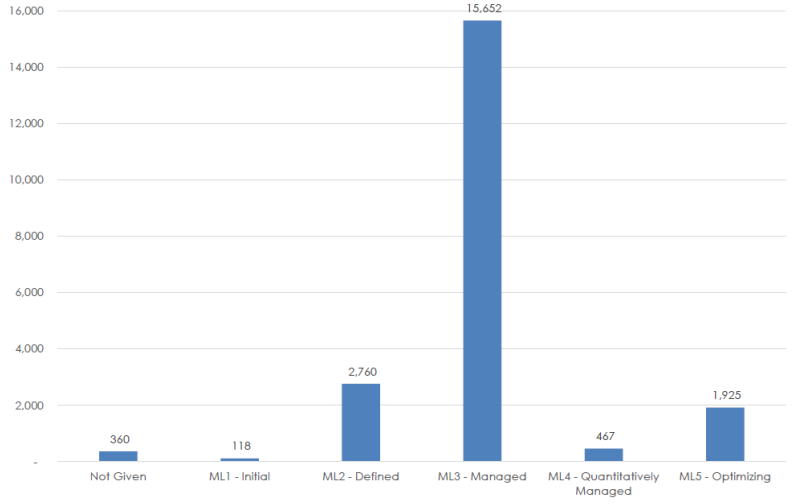
https://cmmiinstitute.com/resource-files/public/cmmi-adoption-trends-2019-mid-year-update

APPRAISALS BY YEAR



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APPRAISALS BY MATURITY LEVEL: TOTALS SINCE 2008

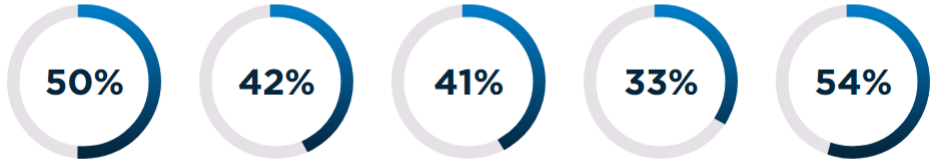


https://www.cmmiinstitute.com/pars/

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How capable is your organization?

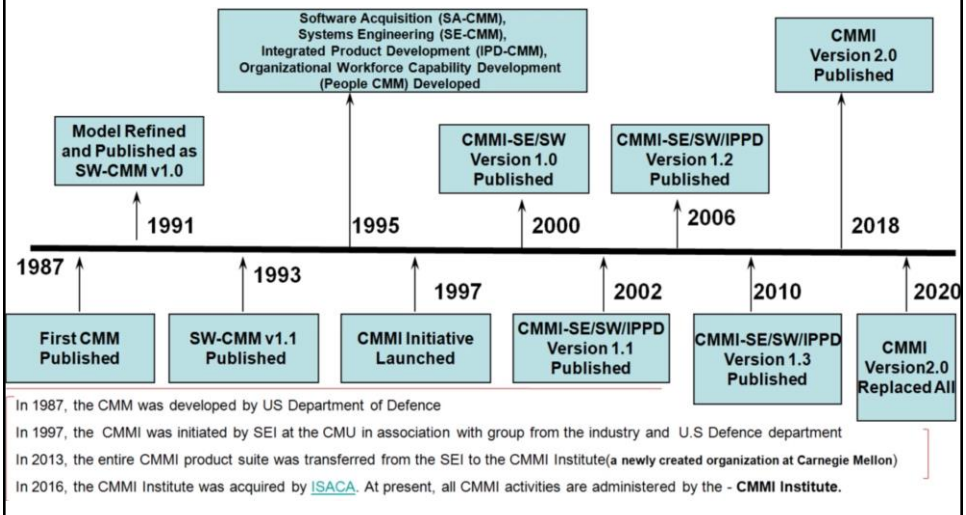
All organizations have capabilities. What many don't have is a process by which they can measure their different capabilities against best practices and pinpoint which ones are driving higher performance.



CMMI Institute

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CMMI timeline



In 1987, the CMM was developed by US Department of Defence

In 1997, the CMMI was initiated by SEI at the CMU in association with group from the industry and U.S Defence department

In 2013, the entire CMMI product suite was transferred from the SEI to the CMMI Institute(**a newly created organization at Carnegie Mellon**)

In 2016, the CMMI Institute was acquired by ISACA. At present, all CMMI activities are administered by the - **CMMI Institute.**

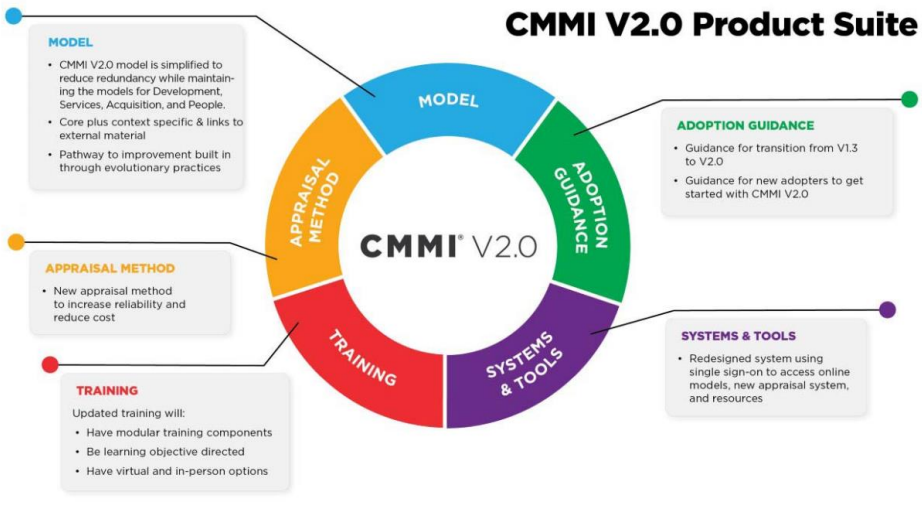
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SEI (Software Engineering Institute)

- Research and development center funded by the US Department of Defense
- Developed at the Carnegie Mellon University
- In 2013 the CMMI Institute was created (http://cmmiinstitute.com/)
- In 2016 the CMMI Institute was acquired by ISACA (an international professional association focused on IT Governance)
- CMMI v1.3 was released in November 2010
- CMMI v2.0 was released in March 2018 – not free!
  - "Help others make measured improvements on their software engineering capabilities"

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CMMI (v2.0)



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CMMI (v2.0) from earlier V1.3

- Focus on **Performance**
  - Performance is key to the success of every organization—no matter their maturity level
  - Level 1 starts by addressing basic performance, and practices from higher levels build on lower levels to further evolve.
- Improved Usability & Integrated Guidance
  - Level 1 practices enable new organizations to take incremental steps to improve and gain early success
  - Includes specific guidance to help organizations that use **agile methods**
- Easier to Understand and Access
  - Written in non-technical business language
- Improved Appraisals
- Adoption Guidelines
  - A clear set of simple steps to get started; List of needed resources;
- New appraisals start on Jan/2019

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CMMI overview (V2.0)

- CMMI stands for Capability Maturity Model Integration
- Process Level Improvement Program
  - Training
  - Implementaion
  - Appraisal
- Reference Model of mature practices
  - Focus on practices/processes
- Maturity = optimization of processes

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CMMI overview (V2.0)

- One model. Multiple (domain) Views:
  - Development
    - Focus on developing quality products and services
    - Products; Components;Services
  - Services
    - Focus to deliver quality services
    - B2B or B2C services
  - Supplier Management
    - Maximize supply chain efficiency
    - Manage suppliers and vendors

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CMMI overview (V2.0)

- CMMI is a group of practice areas (process areas in earlier versions)
- Each Practice area is adapted to the company based on:
  - Culture
  - Behaviours
- Explain the “what” we need and not the “how”
- CMMI is not a standard (ISO).
  - In CMMI you will achieve a “rating” not a certification.

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CMMI overview (V2.0)

- Organizations are appraised and rated at Capability or Maturity Level

CMMI Concepts (V2.0)

- Maturity Model:
  - Structure that describes behaviours, practices and processes to achieve better SW
- Provides:
  - Common language
  - A framework
  - Definition of improvement

CMMI Concepts (V2.0)

- Practices
  - Describe infrastructure or activities that ensure implementation of a practice area
- Practice Area
  - Group of related practices in a specific área
  - When inplemented, make an improvement in that area
  - Return an Intent and Value to achieve a certain capability
- Capability Area
  - Group of related and logical practice area

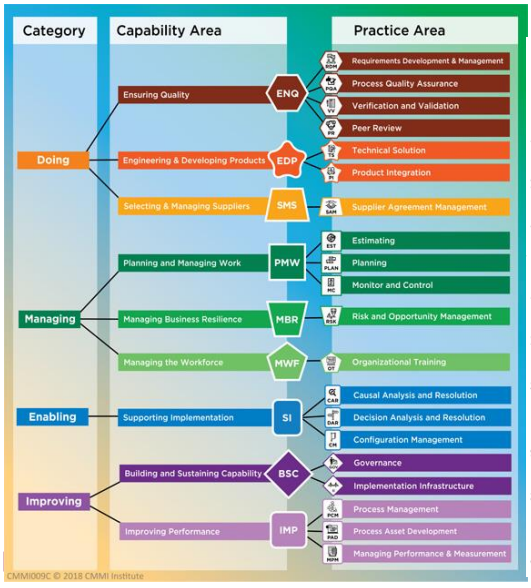
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CMMI Concepts (V2.0)

- Category Area
  - Logical group of capability areas
- CMMI model V2.0:
  - 4 Category Areas
  - 12 Capability Areas
  - 25 Practice Areas

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Capability areas for DEV View



- Project-level
- Project-level
- Project-level
- Project-level
- Project-level
- Org-level
- Project-level
- Across the board
- Org-level*

Category Area

Category Area	focus on
DOING	focus is on ▪ producing ... ▪ developing and delivering of quality solutions, products or services
MANAGING	relates to planning and managing part . This focus on ▪ implementation of solutions ▪ coordinating the work , also the workforce
ENABLING	focus on Enabling part of work , through.. ▪ communicating stakeholders, making decisions ▪ integrating work products and people ▪ analyzing the causes
IMPROVING	focus is on ▪ improving and sustaining the organizational performance

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CMMI Concepts (V2.0)

- Each practice area:
  - Has an intent and value statements
- Each Practice may have Value statement.

- Intent provides process Importance
- Value provides business impact

- Each Practice is indexed by the refering level and order inside the practice area.
  - E.G. *CAR 2.1 Select outcomes for analysis.*
    - First practice related with Maturity Level 2, of practice area "CAR - Causal Analysis".

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Practices (example) –(V2.0)

Category: ENABLING, ***Capability Area***: Supporting Implementation [SI]
CAR CAUSAL ANALYSIS AND RESOLUTION (CAR)

CAR **Intent:** Identify causes of selected outcomes and take action to either prevent recurrence of undesirable outcomes or ensure recurrence of positive outcomes.
CAR **Value**: Addressing root cause issues eliminates rework and directly improves quality and productivity.

CAR 1.1 Identify and address causes of selected outcomes.

CAR 2.1 Select outcomes for analysis.
CAR 2.2 Analyze and address causes of outcomes.

CAR 3.1 Determine root causes of selected outcomes by following an organizational process.
CAR 3.2 Propose actions to address identified root causes.
CAR 3.3 Implement selected action proposals.
CAR 3.4 Record root cause analysis and resolution data.
CAR 3.5 Submit improvement proposals for changes proven to be effective.

CAR 4.1 Perform root cause analysis of selected outcomes using statistical and other quantitative techniques.
CAR 4.2 Evaluate the effect of implemented actions on process performance using statistical and other quantitative techniques.

CAR 5.1 Use statistical and other quantitative techniques to evaluate other solutions and processes to determine if the resolution should be applied on a broader scale.

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Practices (example) –(V2.0)

REQUIREMENTS DEVELOPMENT AND MANAGEMENT (RDM)

Elicit requirements, ensure common understanding by stakeholders, and align requirements, plans, and work products.

Level 1

RDM 1.1	Record requirements.

Level 2

RDM 2.1	Elicit stakeholder needs, expectations, constraints, and interfaces or connections.
RDM 2.2	Transform stakeholder needs, expectations, constraints, and interfaces or connections into prioritized customer requirements.

...

Level 3

RDM 3.1	Develop and keep requirements updated for the solution and its components.
RDM 3.2	Develop operational concepts and scenarios.

...

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Practices (example) –(V2.0)

- For the Requirements Management **Practice Area**:
  - An example **Goal** (required):
    - "Manage Requirements"
  - An example **Practice** to support the Goal (required):
    - "Maintain bi-directional traceability of requirements"
  - Examples (suggested, but not required) of typical **Work Products** might be
    - Requirements traceability matrix or
    - Requirements tracking system

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CMMI Levels (v2.0)

- Level is a representation of what goals (practices have been accomplished)
- Numeric representation of the practices implemented
- Level can be represented as Capability Level or Maturity Level

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CMMI Levels (v2.0)

Maturity Level (ML)	Capability Level (CL)
Achievement of organizations's process improvement applied to multiple practice áreas	Achievement of organizations' practices applied to an individual area
Six maturity Levels (0 to 5)	Four capability levels (0 to 3)
Improvement measured in MLs	Improvement measured in CLs

https://cmmiinstitute.com/learning/appraisals/levels

- After Company appraisal organization achieve a ML or CL (e.g. for ML 4):



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Capability Level 0: Incomplete

Incomplete approach to meeting the intent of the Practice Area.

May or may not be meeting the intent of any practice.

Inconsistent performance.

Capability Level 1: Initial

Initial approach to meeting the intent of the Practice Area.

Not a complete set of practices to meeting the full intent of the Practice Area.

Addresses performance issues.

Capability Level 2: Managed

Subsumes level 1 practices.

Simple, but complete set of practices that address the full intent of the Practice Area.

Does not require the use of the organizational assets.

Identifies and monitors progress towards project performance objectives.

Capability Level 3: Defined

Builds on level 2 practices.

Uses organizational standards and tailoring to address project and work characteristics.

Projects use and contribute to organization assets.

Focuses on achieving both project and organizational performance objectives.

Maturity Level 0: Incomplete
Ad hoc and unknown. Work may or may not get completed.

Maturity Level 1: Initial
Unpredictable and reactive. Work gets completed but is often delayed and over budget.

Maturity Level 2: Managed
Managed on the project level. Projects are planned, performed, measured, and controlled.

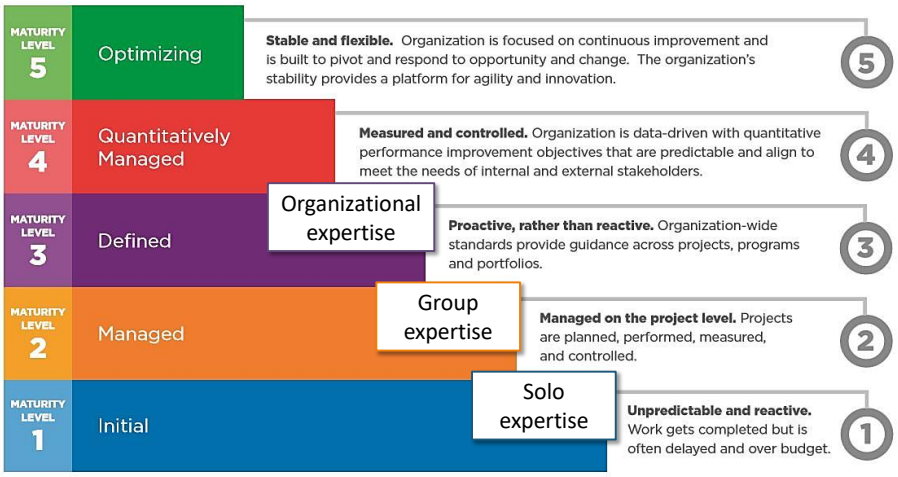
Maturity Level 3: Defined
Proactive, rather than reactive. Organization-wide standards provide guidance across projects, programs, and portfolios.

Maturity Level 4: Quantitatively Managed
Measured and controlled. Organization is data-driven with quantitative performance improvement objectives that are predictable and align to meet the needs of internal and external stakeholders.

Maturity Level 5: Optimizing
Stable and flexible. Organization is focused on continuous improvement and is built to pivot and respond to opportunity and change. The organization's stability provides a platform for agility and innovation.

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Staged Maturity levels



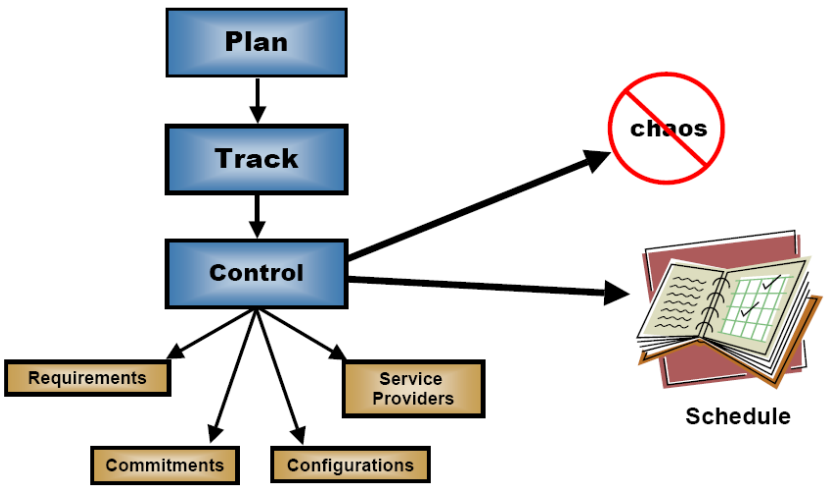
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Level 1 (Initial)

- Organization has no processes
- Any success is probably attributed to a strong and experienced leader
- The probability of repeating this success is low

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Level 2 (Managed) vision



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Level 2 (Managed) - Benefits

- Organization has mastered key project management related processes and is expected to be able to repeat its success when given similar processes
- Managers:
  - take commitments seriously
  - protect sound development practices
  - control baselines
- Developers:
  - have time to do a professional job
  - can meet schedules
  - morale increases

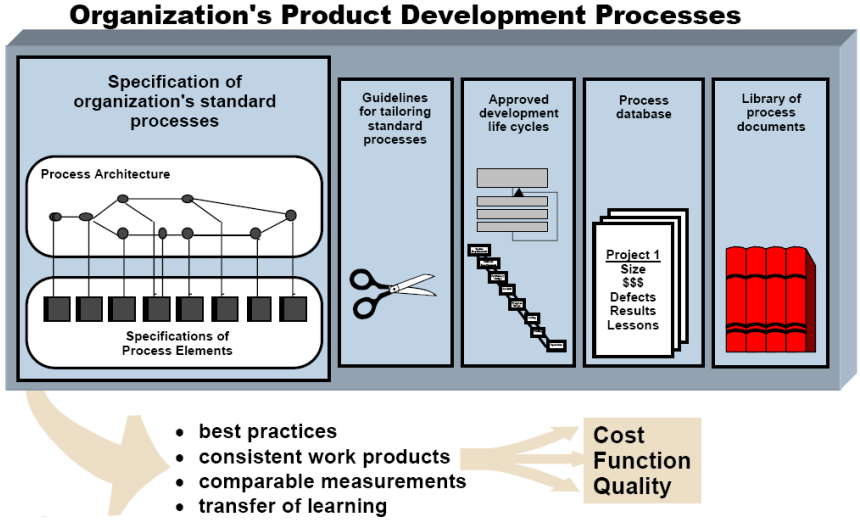
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Level 2 (Managed) – Practice Areas

- CM - Configuration Management
- MA - Measurement and Analysis
- PMC - Project Monitoring and Control
- PP - Project Planning
- PPQA - Process and Product Quality Assurance
- REQM - Requirements Management
- SAM - Supplier Agreement Management

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Level 3 (Defined) - Vision



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Level 3 (Defined) - Benefits

- Organization has mastered major processes related to construction of software along with additional project management related processes
- Managers:
  - organization supports managers
  - estimates are more accurate
  - cost, schedule, and functionality in control
  - quality improves
- Developers:
  - common foundation for professional work
  - foundation for technology benefits
  - transfer among projects is enhanced
  - a common engineering culture emerges

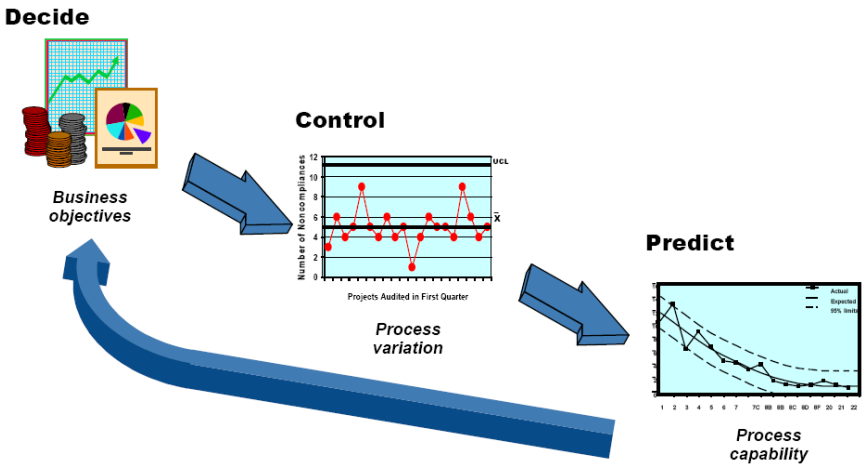
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Level 3 (Defined) – Practice Areas

- DAR - Decision Analysis and Resolution
- IPM - Integrated Project Management +IPPD
- OPD - Organizational Process Definition +IPPD
- OPF - Organizational Process Focus
- OT - Organizational Training
- PI - Product Integration
- RD - Requirements Development
- RSKM - Risk Management
- TS - Technical Solution
- VAL - Validation
- VER - Verification

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Level 4 (Quantitatively Managed) – Vision



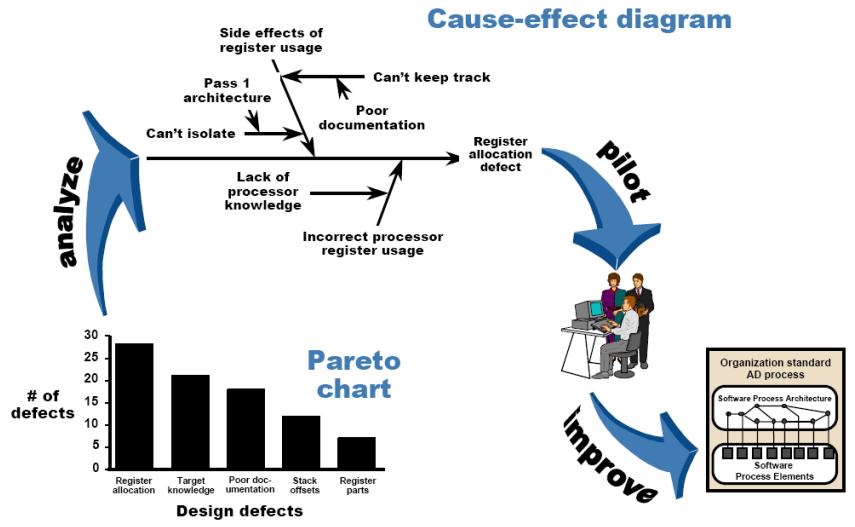
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Level 4 (Quantitatively Managed) - Benefits

- Benefits:
  - Organization focuses its effort on quantitative and quality management
- Practice Areas:
  - QPM - Quantitative Project Management
  - OPP - Organizational Process Performance

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Level 5 (Optimizing) - Vision



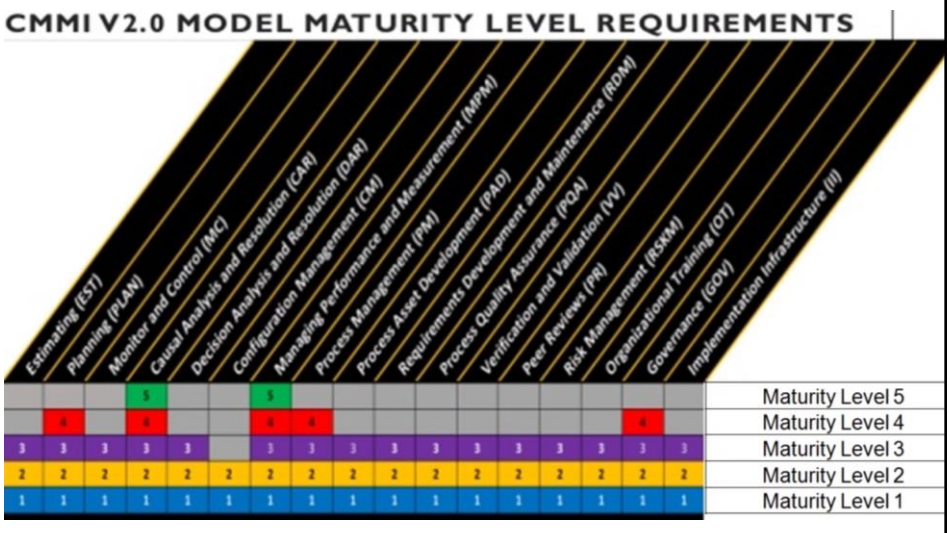
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Level 5 (Optimizing) - Benefits

- Benefits:
  - Organization focuses on continuous improvement
- Practice Areas:
  - CAR - Causal Analysis and Resolution
  - OID - Organizational Innovation and Deployment

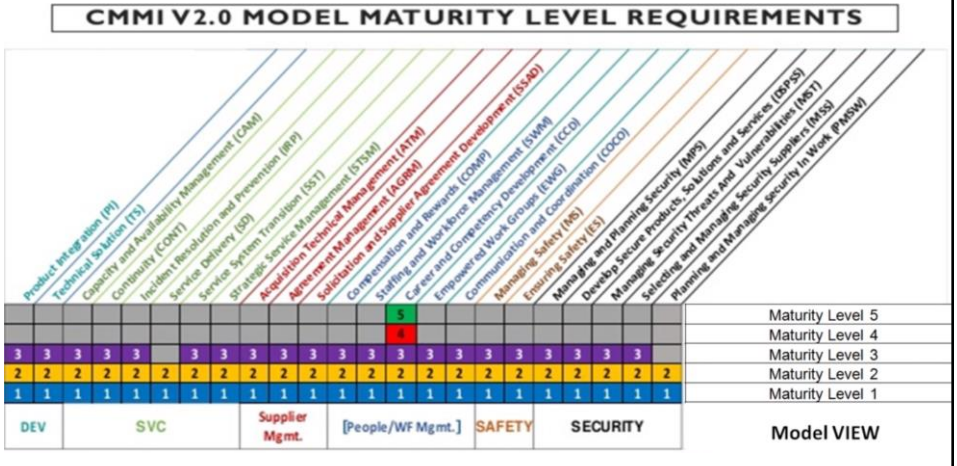
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CORE Areas



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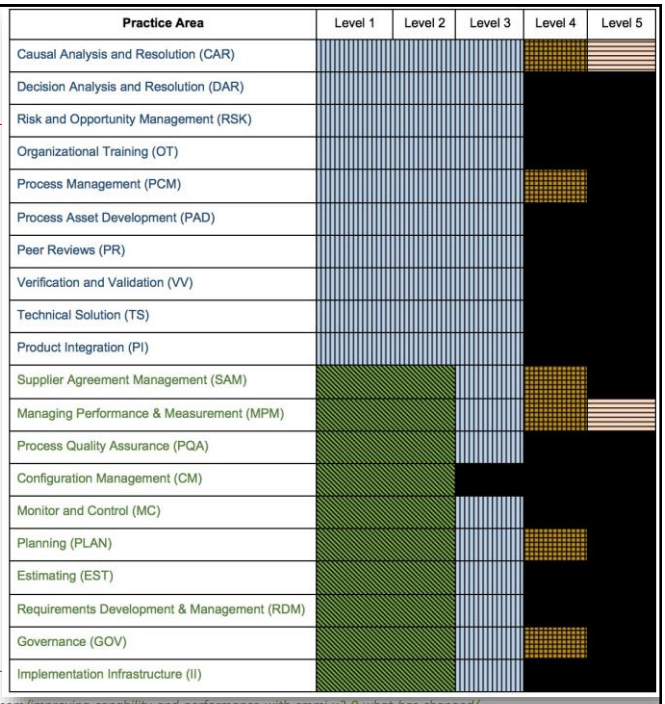
Specific Areas



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Maturity Levels VS Practice Areas

- Level 2
- Level 3
- Level 4
- Level 5



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- *http://processgroup.com/improving-capability-and-performance-with-cmmi-v2-0-what-has-changed/*

V2.0 versus V1.3 - Some changes

- The **term practice area** will be used in place of **process area**.
- **Estimation** will be added as an exclusive and separate practice area.
- Generic goals and generic practices are replaced by some new **practice areas**.
- The 3 constellations (DEV, SVC and ACQ) integrated in a single model.
- Structure of the framework (5 layers):
  - 4 **Categories** - Doing, Managing, Enabling and Improving.
  - 9 **Capability Areas**
  - 20 **Practice Areas**
  - 5 **Capability Levels**
  - 198 **Practices**
    - https://business-process-improvement-blog.blogspot.com/2017/09/cmmi-v20-some-major-changes-and.html
    - https://business-process-improvement-blog.blogspot.com/2018/04/cmmi-v20-some-interesting-observations.html

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Representation Models (V1.3)

- Staged
  - Better applied to assessing the maturity of an organization
  - Provides a roadmap for implementing
    - groups of process areas
    - sequencing of implementation
- Continuous
  - More applicable to the assessment and improvement of processes
  - Provides maximum flexibility for focusing on specific process areas according to business goals and objectives

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Representation Models (V1.3)

- Staged
  - Process improvement is measured using **maturity** levels.
  - Maturity level is the degree of process improvement across a **predefined set of process areas**.
  - Organizational maturity pertains to the "maturity" of a **set of processes** across an organization
- Continuous
  - Process improvement is measured using **capability** levels.
  - Capability level is the achievement of process improvement within an **individual process area**.
  - Process area capability pertains to the "maturity" of a **particular process** across an organization.

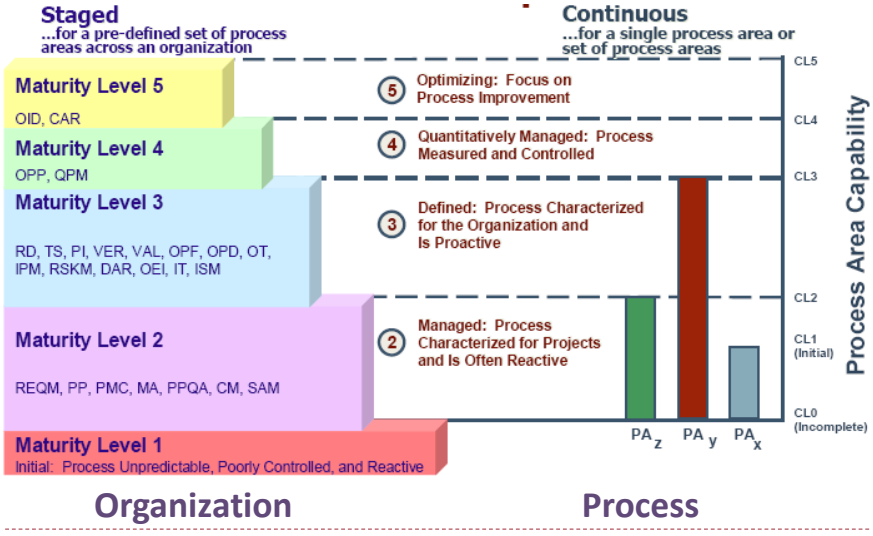
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CMMI (v1.3)

- Is a framework of management, engineering, and support best practices organized into 22 topics, called Process Areas
- Four areas of interest:
  - Product and service **development** — CMMI for Development (CMMI-DEV)
  - **Service** establishment, management, and delivery — CMMI for Services (CMMI-SVC)
  - Product and service **acquisition** — CMMI for Acquisition (CMMI-ACQ)
  - Improving the capability of an organization's **workforce** – People Capability Maturity Model (P-CMM)
- Is an approach to improving the capability of any of the 22 Process Areas by incrementally applying a set of Generic Practices that enhance the functioning of an individual process
- Is best thought of as a set of "process requirements" that help guide an organization who is defining and implementing processes related to the 22 topics
- Is NOT a pre-defined, implementable "as is" process definition

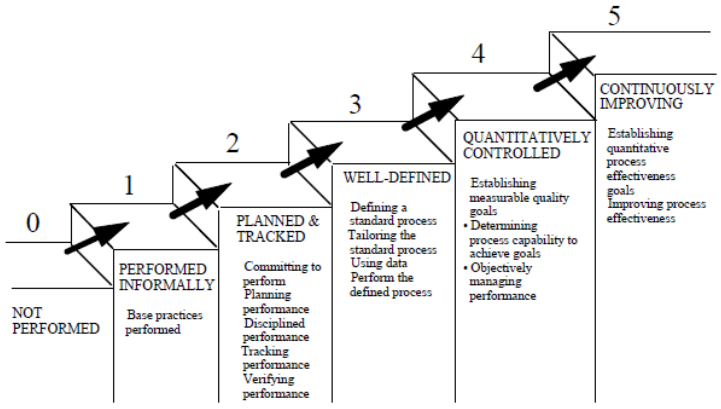
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Representation Models (V1.3)



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CMMI **Continuous** Representation (V1.3)



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Process Areas of CMMI Continuous Representation / Capability Level (V1.3)

Process Area Categories	Process Area	
Process Management	•Organizational Process Focus •Organizational Process Definition + IPPD •Organizational Training	•Organizational Process Performance •Organizational Innovation and Deployment
Project Management	•Project Planning •Project Monitoring and Control •Supplier Agreement Management	•Risk Management •Quantitative Project Management •Integrated Project Management + IPPD
Engineering	•Requirement Management •Requirements Development •Technical Solution	•Product Integration •Verification •Validation
Support	•Configuration Management •Process and Product Quality Assurance •Measurement and Analysis	•Decision Analysis and Resolution •Causal Analysis and Resolution

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Process Areas of CMMI Staged / Maturity Level (V1.3)

Maturity level	Focus	Practice Areas	
5. Optimizing	Continuous Process Improvement	•Organization Innovation and Deployment •Causal Analysis and Resolution	
4. Quantitatively Managed	Quantitative Management	•Organizational Process Performance •Quantitative Project Management	
3. Defined	Process Standardization	•Requirements Development •Technical Solution •Product Integration •Verification •Validation •Organizational Process Focus •Organizational Process Definition	•Integrated Project Management •Risk Management •Decision Analysis and Resolution •Organizational Environment for Integration •Integrated Teaming •Integrated Supplier Management •Organizational Training
2. Managed	Basic Project Management	•Requirements Management •Project Monitoring and Control •Supplier Agreement Management	•Project Planning •Measurement and Analysis •Product and Process Quality Assurance •Configuration Management
1. Initial			

SPICE

- Software Process Improvement and Capability Determination also known as ISO/IEC 15504
- Since 2015, **ISO/IEC 33001** *Information technology -- Process assessment*

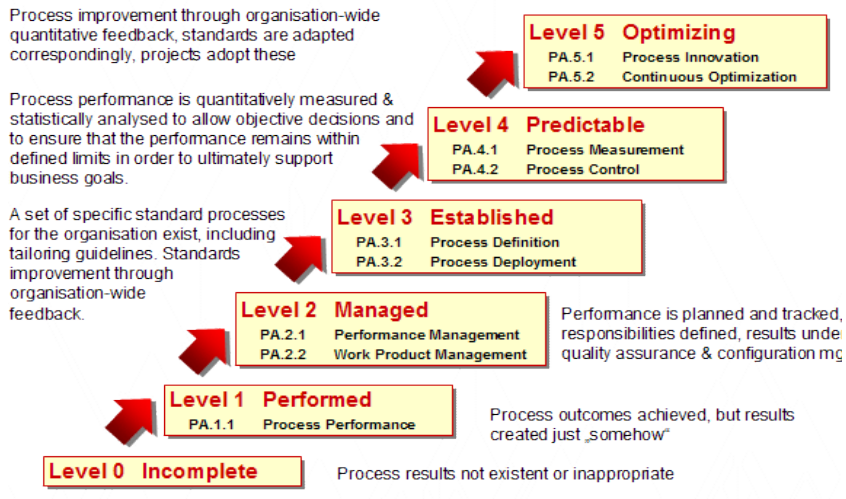


SPICE - Software Process Improvement and Capability Determination

- Extension of the **ISO/IEC 12207**
- Used in two contexts:
  - Capability determination (supplier's evaluation)
    - Created to **assess** the organization's capability for developing software products, systems, or IT services
  - Process Improvement (organization's evaluation)
    - Provides a **Reference Model** (based on ISO/IEC 12207) with a *process dimension* and a *capability dimension,* for defining, measuring and improving the capability of processes such as software development, project management, and quality assurance, among others.
  - For each process, defines a capability level on the scale 0-5
    - process attributes (PA) are used for measuring the capability of the processes
    - Each process attribute is assessed on a four-point rating scale: Not achieved (<15%), Partially achieved (>15% - 50%), Largely achieved (>50%- 85%), Fully achieved (>85%)

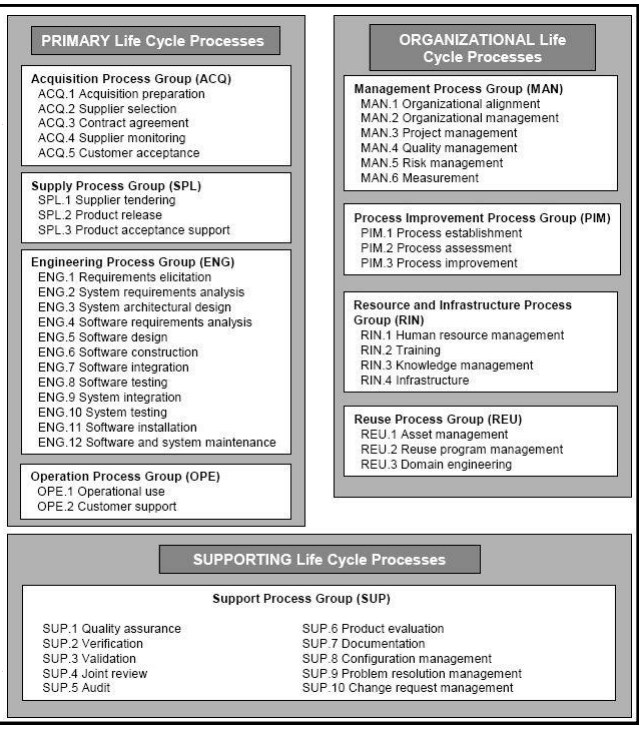
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Capability levels



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Process areas (ISO/IEC 15504)



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Domain specific models

- Automotive SPICE
  - Used to perform conformant assessments of the software process capability of automotive
- SPICE 4 SPACE
  - Assessment model incorporating space software practices following ECSS (European Cooperation for Space Standardization) standard requirements for the production of space software
- Medi SPICE
  - Aims to define a conformity assessment approach for the medical device industry
- Enterprise SPICE
  - Brings together processes from other widely accepted models and standards into an integrated approach for assessment and improvement thereby alleviating the need to use otherwise multiple assessment models.

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Automotive SPICE

- Published in 2005 by the **Special Interest Group Automotive**, an industry-specific standard
- The goal was to adapt SPICE to the specific requirements of the **automobile** manufacturer in order to be able to assess their suppliers in a standard way
  - Has its own **Process Reference Model** based on **ISO/IEC 12207 AMD1 2002 and AMD2 2004**, and **Process Assessment Model**, based on the capability levels from **ISO/IEC 15504**
  - 21 processes have been removed and 5 processes added from the **ISO 12207** model. The remaining processes, for the most part, represented a detailed description of existing professional practices.
  - The German automobile manufacturers Audi, BMW, DaimlerChrysler and VW agreed on a minimal subset of 15 processes, called **Hersteller Initiative Software (HIS) Scope**, which will be assessed by each manufacturer.

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ISO/IEC 15504 and Automotive Spice – Process dimension

Management Process Group (MAN)	Engineering Process Group (ENG)	Supporting Process Group (SUP)
MAN.1 Organizational alignment	A ENG.1 Requirements elicitation	A SUP.1 Quality assurance
MAN.2 Organization management	A ENG.2 System requirements analysis	A SUP.2 Verification
A MAN.3 Project management	A ENG.3 System architectural design	SUP.3 Validation
MAN.4 Quality management	A ENG.4 Software requirements analysis	A SUP.4 Joint review
A MAN.5 Risk management	A ENG.5 Software design	SUP.5 Audit
A MAN.6 Measurement	A ENG.6 Software construction	SUP.6 Product evaluation
	A ENG.7 Software integration	A SUP.7 Documentation
	A ENG.8 Software testing	A SUP.8 Configuration management
	A ENG.9 System integration	A SUP.9 Problem resolution management
	A ENG.10 System testing	A SUP.10 Change request management
	ENG.11 Software installation	
	ENG.12 Software and system maintenance	
The Acquisition Process Group (ACQ)	**Resource & Infrastructure Process Group (RIN)**	**Operation Process Group (OPE)**
ACQ.1 Acquisition preparation	RIN.1 Human resource management	OPE.1 Operational use
ACQ.2 Supplier selection	RIN.2 Training	OPE.2 Customer support
A ACQ.3 Contract agreement	RIN.3 Knowledge management	
A ACQ.4 Supplier monitoring	RIN.4 Infrastructure	
ACQ.5 Customer acceptance		
A ACQ.11 Technical requirements		
A ACQ.12 Legal and administrative requirements		
A ACQ.13 Project requirements		
A ACQ.14 Request for proposals		
A ACQ.15 Supplier qualification		
Supply Process Group (SPL)	**Process Improvement Process Group**	**Reuse Process Group (REU)**
A SPL.1 Supplier tendering	PIM.1 Process establishment	REU.1 Asset management
A SPL.2 Product release	PIM.2 Process assessment	A REU.2 Reuse program management
SPL.3 Product acceptance support	A PIM.3 Process improvement	REU.3 Domain engineering

A Automotive-SPICE | new HIS-Scope | not included in ISO/IEC IS 15504-5

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Automotive SPICE process reference model

